

ME COMPUTER SCIENCE AND ENGINEERING 1st YEAR 1st SEM. EXAM.-2026

Subject: Cloud Computing

Time: 3 hours

Full Marks: 100

The figures in the margin indicate full marks

Candidates are required to give their answers in their own words as far as practicable

Answer Question 1 and **any four** from the rest

1. Answer any ten of the following questions: [10 × 2]
- a) What do you mean by scalability in a distributed system?
 - b) Under what condition can an operation $f(x)$ be considered non-idempotent?
 - c) Which call semantic is implemented when the server replies only after executing an operation and the client reissues the request only if no reply is received within a predefined time? Explain.
 - d) What is marshalling in Remote Procedure Call (RPC)?
 - e) The term "Cloud" is used as a metaphor for the Internet. Justify.
 - f) Your company plans to launch a global web application using AWS infrastructure. Which TWO benefits of AWS Global Infrastructure are most relevant? (Cost savings, Fault tolerance, High availability, Regulatory compliance)
 - g) How does x86 architecture handle an attempt to execute a Ring 0 instruction from a Ring 3 user process?
 - h) Why is cloud computing considered cost-effective for startups? Provide two reasons.
 - i) If a company requires maximum control over its infrastructure, which cloud service model—IaaS, PaaS, or SaaS—is most appropriate? Justify.
 - j) A government agency wants complete control over its IT infrastructure but intends to use AWS for certain applications. Which cloud deployment model is most suitable? Justify.
 - k) An organization must keep sensitive data on-premises for compliance but uses cloud resources to scale during peak demand. Which deployment model does this represent? Justify.
 - l) Why is data redundancy important in cloud storage systems?

[Turn over

- 2.a) What is a Distributed Hash Table (DHT)? Explain how Chord implements a scalable DHT with an example. Show that the number of hops required to look up a key is independent of the number of nodes in the system. [2+5+3]
- b) How can server crashes and client crashes be handled in an RPC system? Discuss strategies to implement at-least-once and at-most-once call semantics. [4+4+2]
- 3.a) Design a leader election algorithm for processes connected in a unidirectional ring with message complexity $O(n \log_2 n)$. Derive the message complexity. Modify the algorithm so that two leaders are elected: one among even-positioned processes and one among odd-positioned processes. [5+2+3]
- b) How can a resource be accessed in a distributed system without knowing its physical location? Design an algorithm that tolerates relocation of the resource while it is being used. [4+3]
- c) Differentiate between idempotent and non-idempotent operations with examples. [3]
- 4.a) Explain different types of Virtual Machine (VM) migration. Discuss Pre-Copy and Post-Copy live migration approaches, including their working mechanisms, advantages, and limitations. [2+5]
- b) A physical host has 2 GB RAM. Is it possible to allocate two VMs each requiring 1.5 GB RAM? Justify using virtualization memory management techniques. [5]
- c) i) Explain how a host assigns multiple IP addresses to different VMs on the same machine.
ii) Describe how a VM sends and receives IP packets in a virtualized network environment. [3+2]
- d) List the managed database services offered by AWS, GCP, and Microsoft Azure. [3]
- 5.a) Explain the working principles of Type-1 and Type-2 hypervisors with examples. [4]
- b) Compare Docker containers and traditional virtual machines in terms of architecture, resource utilization, and isolation. [4]
- c) Explain the need for container orchestration and why Kubernetes is widely used. Describe Kubernetes architecture, focusing on Pods and Kubelets. [2+6]
- d) Identify fully managed AWS services for running containers and Kubernetes workloads. [4]

- 6.a) Is HDFS suitable for storing large graphs? Justify with an example. [3]
- b) What is eventual consistency? How can it be implemented in HDFS? [2+3]
- c) A social graph contains users as nodes and friendships as edges. Explain how such a graph can be stored in HDFS and write MapReduce pseudocode to find users with the highest number of friends. [2+5]
- d) Describe an iterative MapReduce approach to compute PageRank. [5]

7. Write short notes on the following: [4 × 5]

- a) YARN Architecture
- b) 0/1 Bin Packing formulation for cloud resource scheduling
- c) HDFS architecture and read/write pipeline
- d) Cloud Security
